



Teyzix Core Internship (June Batch)

Task ID: DA-3

Task Title: Smart Public Transport Analytics & Business Intelligence Dashboard

Domain: Data Analytics

Difficulty Level: Advanced (Industry-Based)

Assigned Date: 30 June 2026

Submission Deadline: 09 July 2026

Estimated Duration: 10 Days

Project Overview

Public transportation systems generate large amounts of operational data every day, including passenger counts, route performance, revenue, delays, vehicle utilization, and service schedules. However, without proper analysis, transport authorities struggle to identify operational inefficiencies, optimize routes, improve passenger satisfaction, and make informed decisions.

Your task is to develop a **Smart Public Transport Analytics & Business Intelligence Dashboard** that transforms raw transportation data into meaningful business insights through data cleaning, analysis, visualization, and interactive dashboards.

Unlike many analytics projects, **applicants are required to design and create their own original dataset**. Public datasets from Kaggle, UCI, GitHub, or any other online source are **strictly prohibited**.

The objective is to demonstrate the complete data analytics lifecycle—from data collection and preparation to dashboard development and business intelligence reporting.

Project Objectives

The objective of this task is to evaluate your ability to solve a real-world business problem by implementing:

- Original Dataset Creation
- Data Collection
- Data Cleaning & Validation
- Exploratory Data Analysis (EDA)
- KPI Development
- Business Intelligence Dashboard
- Data Visualization
- Business Insights
- Report Generation
- Decision-Making Recommendations

Core Requirements

1. Dataset Creation (Mandatory)

Applicants **must create their own original dataset**.

The dataset should contain at least:

- **500 records**
- **12 or more meaningful features**

Possible features include:

- Route ID
- Route Name
- Bus Number
- Bus Type
- Date

- Departure Time
- Arrival Time
- Delay Duration
- Passenger Count
- Ticket Revenue
- Fuel Consumption
- Driver ID
- Weather Condition
- Day Type (Weekday/Weekend)
- Trip Status

Applicants may create the dataset through:

- Manual Data Collection
- Field Observation
- Simulated Transport Operations
- Realistic Synthetic Data Generation

2. Data Cleaning & Validation

Perform:

- Missing Value Handling
- Duplicate Removal
- Data Validation
- Standardization
- Outlier Detection
- Data Type Correction

3. Exploratory Data Analysis (EDA)

Perform comprehensive analysis including:

- Statistical Summary
- Trend Analysis
- Correlation Analysis
- Passenger Distribution
- Revenue Distribution
- Delay Analysis
- Route Performance Analysis
- Visual Data Exploration

Provide meaningful observations and business insights.

4. KPI Development

Create important business KPIs such as:

- Total Passengers
- Total Revenue
- Average Daily Passengers
- Average Delay Time
- Most Profitable Routes
- Least Performing Routes
- Peak Travel Hours
- Average Occupancy Rate
- Daily Trip Completion Rate

5. Interactive Dashboard

Develop a professional dashboard displaying:

- Passenger Trends
- Revenue Analysis
- Route Performance
- Delay Analysis
- Peak Travel Hours
- Fuel Consumption
- Route Comparison
- Interactive Filters
- Executive Summary

Applicants may use:

- Power BI
- Tableau
- Excel Dashboard
- Looker Studio

6. Business Insights

Based on your analysis, identify:

- Highest Revenue Routes
- Most Crowded Routes
- Peak Passenger Hours
- Delay Hotspots

- Low Performing Routes
- Operational Inefficiencies
- Resource Utilization
- Passenger Demand Trends

7. Recommendations

Provide actionable recommendations for transport authorities, such as:

- Route Optimization
- Schedule Improvements
- Peak Hour Fleet Planning
- Revenue Optimization
- Delay Reduction Strategies
- Fuel Efficiency Improvements
- Passenger Experience Enhancements

8. Report Generation

Prepare a professional analytics report including:

- Executive Summary
- Data Cleaning Summary
- Key Performance Indicators
- Dashboard Screenshots
- Business Insights
- Recommendations
- Conclusion

9. Documentation

Clearly document:

- Dataset Creation Method
- Data Dictionary
- Data Cleaning Process
- Dashboard Development
- KPI Definitions
- Challenges Faced
- Future Improvements

Bonus Features (Optional)

Applicants may implement:

- SQL Database Integration
- Python-based Data Analysis
- Interactive Drill-Down Reports
- Geographic Route Mapping
- Real-Time Dashboard Simulation
- Automated Report Generation
- Mobile-Friendly Dashboard
- Power BI Service Publishing

Technical Requirements

Your project should demonstrate:

- Original Dataset Design
- Data Cleaning
- Data Validation
- Exploratory Data Analysis
- Business Intelligence
- Dashboard Development
- Data Visualization
- KPI Design
- Report Writing
- Professional Documentation

Recommended Technology Stack

Data Analysis

- Microsoft Excel
- SQL
- Python (*Pandas, NumPy*) (*Optional*)

Dashboard Development

- Power BI
- Tableau

- Looker Studio

Visualization

- Power BI Visuals
- Tableau Charts
- Excel Charts
- Python (*Matplotlib / Plotly*) (*Optional*)

Submission Requirements

Applicants must submit:

- Original Dataset (CSV/Excel)
- Power BI / Tableau Dashboard File
- Dashboard PDF Export
- Business Analytics Report
- Data Dictionary
- README.md
- GitHub Repository (*Optional but Recommended*)
- Screenshots
- ZIP File

Evaluation Criteria

Submissions will be evaluated based on the following criteria:

- **Dataset Quality & Originality (20%)** – Relevance, completeness, and originality of the self-created dataset.
- **Data Cleaning & Preparation (15%)** – Effective preprocessing, validation, and organization of the dataset.
- **Dashboard Design & Visualization (20%)** – Interactive, professional, and easy-to-understand dashboards with meaningful visualizations.
- **Business Insights & KPI Analysis (20%)** – Ability to extract actionable insights and develop relevant business KPIs.
- **Report Quality & Recommendations (10%)** – Clear reporting and practical recommendations based on data analysis.
- **Documentation & File Organization (10%)** – Proper documentation, data dictionary, and project organization.
- **Innovation & Creativity (5%)** – Additional analytics, visualizations, or business intelligence features that add value.

Submission Guidelines

- Creating your own dataset is **mandatory**.
- Public datasets from Kaggle, UCI, GitHub, or any other online source are **strictly prohibited**.
- Clearly explain how the dataset was collected or generated.
- Ensure the dataset is realistic, well-structured, and suitable for business analysis.
- Design dashboards that are interactive, professional, and easy to understand.
- Provide meaningful business insights rather than only charts and graphs.
- Plagiarism of datasets, dashboards, or reports will result in immediate disqualification.

Expected Learning Outcomes

By completing this task, interns will gain practical experience in:

- Dataset Design & Collection
- Data Cleaning & Preparation
- Exploratory Data Analysis (EDA)
- Business Intelligence
- Dashboard Development
- KPI Design
- Data Visualization
- Business Reporting
- Decision Support Systems
- Industry-Standard Data Analytics Workflow

Note

This project simulates a real-world analytics solution used by transport authorities and smart city organizations to monitor public transportation performance, improve operational efficiency, optimize routes, and enhance passenger experience. Applicants are expected to demonstrate originality by creating their own dataset and applying professional data analytics practices. Exceptional submissions may be shortlisted for inclusion in the **TEYZIX CORE** portfolio as showcase projects.
